

# Mohamad Ali Al Saidi

AI & Machine Learning Engineer

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LinkedIn | GitHub | Portfolio

## Professional Summary

AI and Machine Learning Engineer and Ph.D. candidate in Artificial Intelligence and Robotics, with hands-on experience in computer vision and OCR, multilingual NLP, model compression, LLM, and TinyML.

Strong software engineering background using Python, Java, PyTorch, TensorFlow, and SQL, with experience taking projects from data preparation and model training through desktop, local, and embedded deployment.

## Technical Skills

**Machine Learning & Deep Learning:** Python, PyTorch, TensorFlow, Keras, Scikit-learn, neural-network training, evaluation, and hyperparameter tuning

**NLP & Large Language Models:** Hugging Face Transformers, BERT, T5, Llama, Mistral, Qwen, NLLB, M2M100, tokenization, embeddings, translation, summarization, RAG, and local LLM inference

**Vision-Language Models:** CLIP, BLIP, LLaVA, multimodal embeddings, and image-text representation

**Computer Vision:** OpenCV, OCR, image preprocessing, object detection, segmentation, YOLOv5, YOLOv8, R-CNN, Fast R-CNN, Faster R-CNN, and Mask R-CNN

**Model Optimization & Edge AI:** Knowledge distillation, model pruning, quantization, TensorFlow Lite, TensorFlow Lite Micro, TinyML, and Arduino Nano 33 BLE

**Software Engineering & Tools:** Java, C/C++, SQL, Git, Linux, Docker, Bash, PowerShell, and LaTeX

## Professional Experience

### Airline System Developer

2018 – 2025

*Sky Gift*

*Beirut, Lebanon*

- Developed and maintained a modular airline reservation and operations platform supporting passenger records, bookings, ticketing workflows, and internal administrative processes.
- Built desktop interfaces and operational dashboards using Java, SQL, and related technologies to improve access to booking, passenger, and flight information.
- Integrated a passport field-detection and OCR workflow into passenger-data processing, combining computer vision, OCR, MRZ validation, and structured data extraction.
- Developed Python and SQL utilities for data processing, validation, reporting, and information exchange between operational systems.
- Supported Windows servers, databases, network infrastructure, backups, access control, and troubleshooting for daily business operations.

### System Administrator Intern

2017 – 2018

*ICT Department, Ferdowsi University of Mashhad*

*Mashhad, Iran*

- Installed, configured, and maintained Windows and Linux workstations, servers, and network equipment across the university ICT environment.
- Assisted with operating-system updates, security patching, account management, backup procedures, and troubleshooting of hardware, software, and connectivity issues.
- Automated recurring administration and maintenance tasks using Bash and PowerShell scripts.

## Teaching & Training Experience

### Machine Learning Course Co-Instructor

Spring 2026

*Ferdowsi University of Mashhad*

*Mashhad, Iran*

- Co-taught a semester-long machine learning course with academic supervisor, sharing responsibility for course delivery throughout the semester.

- Delivered a two-day live workshop series for teachers on the practical and responsible use of AI in education, covering prompt design, lesson planning, assessment materials, presentations, and classroom content creation.

## Selected Projects

### Passport Field Detection and OCR | YOLOv8, EasyOCR, Tesseract, OpenCV, Python

- Trained a YOLOv8n model on a custom dataset of 604 passport images to detect seven fields: MRZ, first name, last name, passport number, birth date, expiry date, and gender.
- Achieved 92.76% mAP@0.5 and 94.26% recall, then integrated field detection with EasyOCR, Tesseract, MRZ check-digit correction, structured output, and a desktop interface.

### Fine-Tuned Local Multilingual Translator | Transformers, NLLB, M2M100, PyTorch, Python

- Fine-tuned multilingual transformer models on a private dataset for local Persian, Arabic, and English translation, producing updated model weights for domain-specific inference.
- Developed an offline desktop application with text segmentation, newline and structure preservation, decoding improvements, model caching, and privacy-preserving local inference.

### Model Pruning and Knowledge Distillation | PyTorch, VGG19, CIFAR-100, Model Compression

- Reproduced and extended a VGG19 compression pipeline on CIFAR-100 using iterative unstructured pruning, recovery training, and knowledge distillation.
- Achieved 39.47% sparsity in the pruned teacher and reduced the smaller student from 40.18M to 16.37M parameters and from 153.35 MB to 62.48 MB while maintaining 73.20% accuracy.

### TinyML Gesture Recognition on Arduino Nano 33 BLE | TensorFlow, CNN, LSTM, TensorFlow Lite Micro, Arduino

- Collected a custom flex-and-wave gesture dataset using six-axis accelerometer and gyroscope measurements from an Arduino Nano 33 BLE.
- Adapted CNN and LSTM pipelines for two-class, six-feature time-series classification, converted and quantized the trained model with TensorFlow Lite, and deployed it for real-time inference on the microcontroller.

## Education

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Ph.D. Candidate in Computer Engineering, Artificial Intelligence &amp; Robotics</b><br><i>Ferdowsi University of Mashhad</i>                                    | <b>2023 – Present</b><br><i>Mashhad, Iran</i> |
| <b>Doctor of Business Administration (DBA)</b><br><i>University of Tehran</i>                                                                                      | <b>2024 – 2026</b><br><i>Tehran, Iran</i>     |
| <b>Master of Business Administration (MBA)</b><br><i>Ferdowsi University of Mashhad</i>                                                                            | <b>2022 – 2023</b><br><i>Mashhad, Iran</i>    |
| <b>M.Sc. in Computer Engineering, Artificial Intelligence &amp; Robotics</b><br><i>Ferdowsi University of Mashhad   GPA: 3.9/4.0</i>                               | <b>2020 – 2022</b><br><i>Mashhad, Iran</i>    |
| – <i>Thesis:</i> Hierarchical Solving of Euclidean Geometry Ruler-and-Compass Construction Problems Using Visual Knowledge Representation and Curriculum Learning. |                                               |
| <b>B.Sc. in Computer Engineering, Artificial Intelligence</b><br><i>Ferdowsi University of Mashhad   GPA: 3.9/4.0</i>                                              | <b>2016 – 2020</b><br><i>Mashhad, Iran</i>    |

## Languages

Arabic: Native | English: Fluent (C1) | Persian: Fluent (C2)